

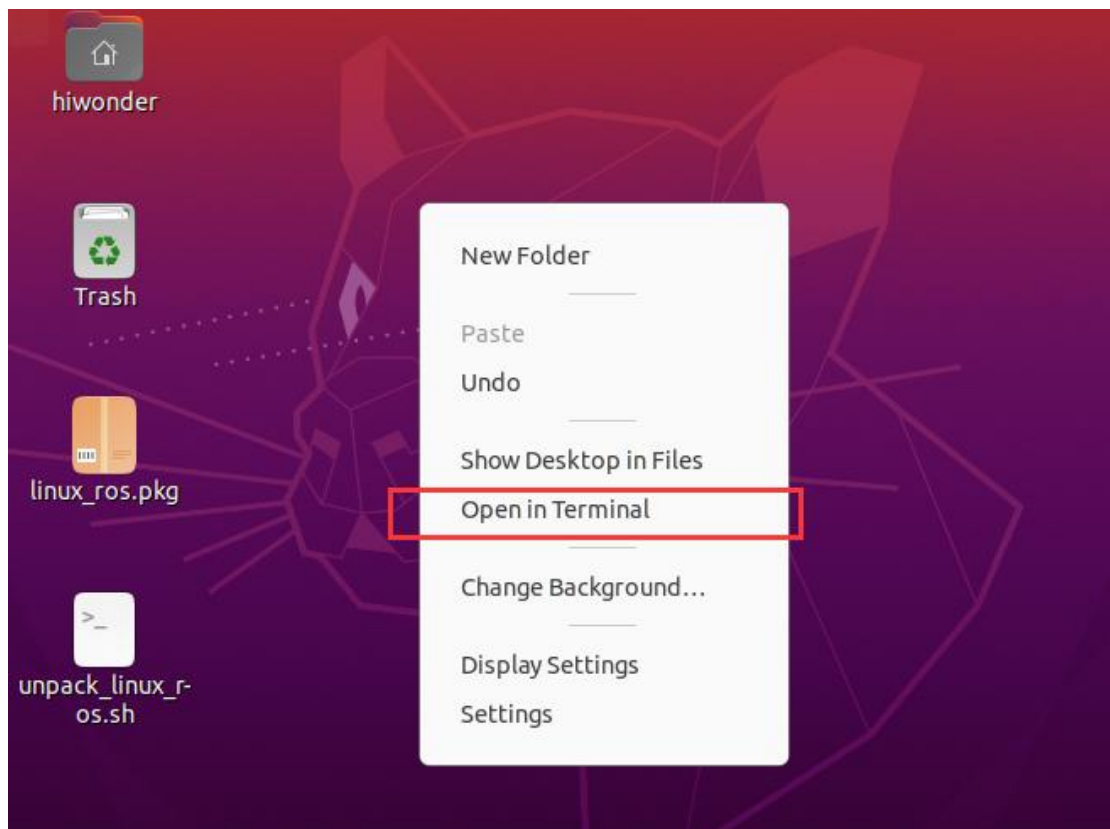
Install Depth Camera ROS SDK

Before starting, make sure to follow the instructions in the “Lesson 1 Depth Camera Configuration” in the same directory to set up the operating environment.

- 1) Copy the “linux_ros.pkg” and “unpack_linux_ros” under the directory of “linux_ros” to the virtual machine.

The virtual machine account username is "hiwonder" and the password is also "hiwonder".

- 2) Right-click the mouse to open the terminal.



- 3) Enter the command to compress the “linux_ros.pkg” package.

```
./unpack_linux_ros.sh
```

```
hiwonder@ubuntu:~/Desktop$ ./unpack_linux_ros.sh
linux_ros/
linux_ros/libs/
linux_ros/libs/include/
linux_ros/libs/include/Logger.h
linux_ros/libs/include/as_camera_sdk_def.h
linux_ros/libs/include/as_camera_sdk_api.h
linux_ros/libs/include/common.h
linux_ros/libs/lib/
linux_ros/libs/lib/arm-linux-gnueabi/
linux_ros/libs/lib/arm-linux-gnueabi/libFilt.so
linux_ros/libs/lib/arm-linux-gnueabi/libasuv.so
linux_ros/libs/lib/arm-linux-gnueabi/libturbojpeg.so
linux_ros/libs/lib/arm-linux-gnueabi/libzmq.so
linux_ros/libs/lib/arm-linux-gnueabi/libAngKondyorArith.so
linux_ros/libs/lib/arm-linux-gnueabi/libasusb.so
linux_ros/libs/lib/arm-linux-gnueabi/libalg_kunlun.so
linux_ros/libs/lib/arm-linux-gnueabi/readme.md
```

- 4) Enter the command to access the function pack.

```
cd linux_ros/ros2/
```

```
hiwonder@ubuntu:~/Desktop$ cd linux_ros/ros2/
```

- 5) Enter the command to copy the ROS package to the ROS environment and press “Enter”.

```
cp -r ascamera build.sh run_ascamera_node.sh /home/hiwonder/ros_ws/
```

```
hiwonder@ubuntu:~/Desktop/linux_ros/ros2$ cp -r ascamera build.sh run_ascamera_node.sh /home/hiwonder/ros_ws/
```

- 6) Enter the command to navigate to the ROS workspace and press “Enter”.

```
cd ./ascamera/scripts/
```

```
hiwonder@ubuntu:~/Desktop/linux_ros/ros2$ cd ./ascamera/scripts/
```

- 7) Enter the command to import the rule file to the workspace. Press “Enter” and input the password “hiwonder”.

```
sudo cp angstrong-camera.rules /etc/udev/rules.d
```

```
hiwonder@ubuntu:~/Desktop/linux_ros/ros2/ascamera/scripts$ sudo cp angstrong-camera.rules /etc/udev/rules.d
[sudo] password for hiwonder:
hiwonder@ubuntu:~/Desktop/linux_ros/ros2/ascamera/scripts$
```

- 8) Enter the command to obtain the target platform for the local GCC compiler. Input the command to make it effective.

```
sudo ./gettarget.sh
```

```
sudo udevadm control --reload-rules
```

```
sudo udevadm trigger
```

```
hiwonder@ubuntu:~/Desktop/linux_ros/ros2/ascamera/scripts$ sudo ./gettarget.sh
x86_64-linux-gnu
hiwonder@ubuntu:~/Desktop/linux_ros/ros2/ascamera/scripts$ sudo udevadm control
--reload-rules
hiwonder@ubuntu:~/Desktop/linux_ros/ros2/ascamera/scripts$ sudo udevadm trigger
```

9) Enter the command to modify the path of the configuration file.

```
gedit ../launch/ascamera.launch.py
```

```
hiwonder@ubuntu:~/ros_ws/ascamera/scripts$ gedit ../launch/ascamera.launch.py
hiwonder@ubuntu:~/ros_ws/ascamera/scripts$
```

/home/hiwonder/ros_ws/ascamera/configurationfiles

```
1 from launch import LaunchDescription
2 import launch_ros.actions
3
4 def generate_launch_description():
5     return LaunchDescription([
6         launch_ros.actions.Node(
7             namespace="ascamera",
8             package='ascamera',
9             executable='ascamera_node',
10            respawn=True,
11            output='both',
12            parameters=[
13                {"confiPath": "/home/hiwonder/ros_ws/ascamera/-
14            configurationfiles"}
15        ],
16    ])
```

10) Run the command to navigate to the workspace.

```
cd ~/ros_ws/
```

```
hiwonder@ubuntu:~/ros_ws/ascamera/scripts$ cd ~/ros_ws/
hiwonder@ubuntu:~/ros_ws$
```

11) Enter the command to recompile the workspace.

```
./build.sh
```

```
hiwonder@ubuntu:~/ros_ws$ ./build.sh
Starting >>> ascamera
Finished <<< ascamera [0.51s]

Summary: 1 package finished [0.90s]
hiwonder@ubuntu:~/ros_ws$
```

12) Enter the command to set the new workspace environment variables.

```
./install/setup.bash
```

```
hiwonder@ubuntu:~/ros_ws$ . ./install/setup.bash
```

13) So far, we have completed the installation of the Depth Camera ROS2 SDK.